

Operating Systems and Networks SoSe 25 Notes

Igor Dimitrov

2024-12-18

Table of contents

Preface	5
I Exam Notes	6
1 Assignment Sheets Overview	7
1.1 Assignment Sheet 1	7
1.2 Sheet 2	8
1.3 Sheet 3	9
1.4 Sheet 4	10
Classification of Übungsblatt 4	10
Topics	18
1.5 Sheet 5	18
Main syllabus-level topics	18
Analysis of Übungsblatt 5	19
Why Aufgabe 2 and Aufgabe 6 stand out	24
Aufgabe 1: important concept, less important packaging	25
Aufgabe 3: TLB capacity + locality	26
Aufgabe 4: TLB and context switches	26
Aufgabe 5: useful, but supplementary	26
Aufgabe 7: very useful calculation pattern	27
Aufgabe 8: multilevel tables	27
My exam-preparation grouping	28
1.6 Sheet 6	29
Main syllabus-level topics	29
Detailed subtopics	29
Assignment-by-assignment classification	29
Recommended study priority	33
1.7 Sheet 7	34
Main syllabus-level topics	34
Detailed subtopics	35
Analysis of the individual assignments	35
Important distinction for Aufgabe 2	39
Recommended exam-preparation priority	39

1.8	Sheet 8	40
	Main syllabus-level topics	40
	Detailed subtopics	40
	Analysis of the individual assignments	41
	Why these tasks are strongly exam-relevant	45
	Recommended preparation priority	46
1.9	Sheet 9	46
	Detailed subtopics	47
	Exercise-by-exercise analysis	47
	Recommended exam-preparation priority	52
1.10	Sheet 10	53
	Main syllabus-level topics	53
	Detailed subtopics	54
	Classification of the individual exercises	54
	Evidence from the reference exams	57
	Recommended study priority	58
1.11	Overview	58
	Strongest sheets overall	61
2	Plan	62
2.1	Priority hierarchy	62
	Priority 1: direct reference-exam matches	62
	Priority 2: likely supporting material	62
	Priority 3: postpone unless time remains	63
2.2	Ten-day core plan	63
2.3	If you have 11–13 days	65
2.4	How to structure each study day	65
2.5	Build an error log, not another long summary	66
2.6	Learn reusable solution templates	66
2.7	The most important rule	67
3	Processes and fork()	68
3.1	Unit 1: <code>fork()</code> , process trees, and process states	68
	1. What is a process?	68
3.2	2. What does <code>fork()</code> do?	69
	Return value	69
	Crucial distinction	70
3.3	3. Counting processes	70
3.4	4. The safe process-tree method	71
	Example	71
3.5	5. Process hierarchy	72
3.6	6. The five basic process states	72
	State transitions	73

Where does <code>fork()</code> fit?	73
3.7 7. Common exam traps	73
4 Exam-like exercise	74
Part A: <code>fork()</code> and the process tree	74
Part B: Process states	75
4.1 Answers	76
Part A	76
Part B	77
5 Race conditions, semaphores and synchronization	78
5.1 Why synchronization is necessary	78
5.2 Race conditions	79
Critical region	79
Requirements of a good mutual-exclusion solution	80
5.3 Atomicity	80
5.4 Semaphores	80
<code>wait(S)</code>	81
<code>signal(S)</code>	81
Binary and counting semaphores	81
5.5 Mutual exclusion with a semaphore	82
Mutex versus binary semaphore	83
5.6 Enforcing an execution order	83
5.7 Blocking versus busy waiting	84
Busy waiting	84
Blocking	84
5.8 Condition variables	84
Why <code>while</code> , not <code>if</code> ?	85
<code>signal</code> versus <code>broadcast</code>	86
Condition variables do not remember notifications	86
Why update and signal under the mutex?	86
Example: producer-consumer queue	87
Complete producer-consumer program	88
Bounded-buffer producer-consumer problem	89
5.9 Producer-consumer pattern	92
5.10 Simplified read-write lock	93
5.11 Common exam traps	95
5.12 Exam-like exercises	95
Exercise 1: Race condition and semaphore repair	95
Exercise 2: Ordering and condition variables	96
Exercise 3: Debugging a semaphore construction	97
5.13 Answers	98
Exercise 1	98

Preface

Part I

Exam Notes

1 Assignment Sheets Overview

1.1 Assignment Sheet 1

Task	Topic / subtopic	Type	Exam relevance	Revision priority
A1	Learning methodology: focused/diffuse thinking, creativity	Research + conceptual / supplementary	Low	Skip for IBN content revision
A2	Computer architecture: x64 general-purpose registers, AVX2 registers, powers of two	Research + calculation	Medium-low	Know the underlying ideas, probably don't spend much time on lookup
A3	OS protection: process address spaces, user/kernel mode, controlled resource access	Conceptual / application	High	Definitely do
A4	Kernel mode / kernel stack; interrupts and system calls	Conceptual / explanation	High	Definitely do
A5	POSIX system calls and minimal interfaces	Conceptual / design	Medium	Useful, but selective

Task	Topic / subtopic	Type	Exam relevance	Revision priority
A6	Bash shell scripting	Programming	Low–medium	Understand shell concepts; full implementation is lower priority
A7	Advantages/disadvantages of abstractions	Conceptual / supplementary	Low	Optional

1.2 Sheet 2

Task	Main topic / subtopic	Problem type	Exam relevance	What I'd do
A1	PCB / Linux <code>task_struct</code>	Research + conceptual	Medium	Learn PCB concept; skip detailed kernel-source research
A2a	<code>fork()</code> , return values, process tree	Trace / code analysis	Very high	Definitely solve fully
A2b	<code>fork()</code> , PID-dependent behavior	Trace + conceptual / exploratory	Medium	Understand, but lower priority
A3	Process hierarchy, process states, <code>ps</code> , background/suspended processes	Practical experiment + conceptual	High	Do conceptually; actual terminal experiment optional
A4	<code>exec*()</code> family	Research / API comparison	Medium-low	Learn core <code>exec</code> semantics; don't memorize all signatures
A5	Process/thread context switching	Conceptual / mechanism	High	Definitely study

Task	Main topic / subtopic	Problem type	Exam relevance	What I'd do
A6.1	Thread creation/join/order	Programming + synchronization	Medium	Useful, but belongs more to Threads
A6.2–3	Thread performance measurement	Empirical / performance	Low	Skip for exam revision

1.3 Sheet 3

Topics: Threads, Race Conditions, Mutual Exclusion, Critical Sections, Semaphores, Mutexes, Condition Variables, Active/Busy Waiting, Producer–Consumer Problem, Read–Write Locks, Deadlock, Starvation, Pthreads synchronization APIs, Win32 synchronization APIs.

Aufgabe	Topic / subtopic	Type of task	Exam relevance	Exam-likeness
1	Concurrency → Race Conditions / Mutual Exclusion	Code analysis, conceptual reasoning	Very high	Very high
2	Synchronization → Semaphores, Mutexes, Condition Variables	Conceptual / video-based / supplementary theory	High	Medium–High
3	Synchronization → Producer–Consumer, Condition Variables, Active Waiting	Programming / implementation / performance experiment	High conceptually, lower for full implementation	Medium

Aufgabe	Topic / subtopic	Type of task	Exam relevance	Exam-likeness
4	Synchronization → Semaphores, Read-Write Locks, Deadlock / Starvation / Mutual Exclusion	Pseudocode analysis + construction	Very high	Very high
5 Bonus	Synchronization APIs → POSIX/Pthreads vs. Win32 Semaphores	Research / API lookup / coding	Low-Medium	Low

1.4 Sheet 4

Yes. Looking at **Übungsblatt 4 as a whole**, there is a fairly clear transition in the course here: Aufgaben 1–3 finish the block on **Interprozesskommunikation / Pipes**, while Aufgaben 4–7 begin **Speicherverwaltung**, especially fragmentation, contiguous allocation, logical/physical addresses and the foundations of paging. This matches the organization of the summary: IPC comes before threads/synchronization, while memory management then introduces swapping, fragmentation, MMU/logical vs. physical addresses, followed by paging. ibn-zusammenfassung ibn-zusammenfassung ibn-zusammenfassung

The exam evidence makes the **memory-management half particularly important**. The 2016 exam devoted an entire 12-point problem to paging and address translation, including page-table sizes, logical/physical addresses, page number/offset and inverted page tables. ibn-klausur-2016 The more recent memory protocol likewise reports a large paging/memory-management problem involving page tables, logical/physical addresses and calculations. Gedächtnisprotokoll_Klausur_IB...

Classification of Übungsblatt 4

Aufgabe	Topic → subtopic	Type of task	Exam relevance	How exam-like?	Assessment
1	Interprocess Communication → named vs. anonymous pipes	Conceptual / application / example construction	Medium	Medium	The concepts themselves are standard OS material and explicitly appear in the summary. You need to understand the limitations and use cases of named vs. anonymous pipes. The requirement to invent your own scenario is somewhat exercise-like rather than typical written-exam style.

Aufgabe	Topic → subtopic	Type of task	Exam relevance	How exam-like?	Assessment
2	IPC / OS interfaces → Win32 handles, I/O redirection	Research + program- ming/API investiga- tion	Low	Very low	Strongly research- oriented: look up Win32 handle repre- sentation and construct Windows API code for redirection. The summary and the available exams concentrate much more on POSIX concepts than on exact Win32 API details. Useful background, but not something I would prioritize for exam preparation.

Aufgabe	Topic → subtopic	Type of task	Exam relevance	How exam-like?	Assessment
3	IPC → anonymous pipes, bidi- rectional process communi- cation	Programming / imple- mentation	Medium	Low– Medium	Anonymous pipes themselves are relevant; the summary explicitly covers <code>pipe()</code> , <code>read()</code> and <code>write()</code> . ibn- zusammenfassung But writing a complete program with two processes and two-way com- munication is more implementation- heavy than the exam evidence suggests. Un- derstanding <i>how</i> the two pipes must be arranged is more important than memorizing code.

Aufgabe	Topic → subtopic	Type of task	Exam relevance	How exam-like?	Assessment
4	Memory manage- ment → fragmenta- tion, swapping, paging, allocation trade-offs	Research / conceptual / diagram interpreta- tion	Medium– High concepts, but Low– Medium as an exercise	Low– Medium	The concepts are important: fragmentation, swapping and paging are core memory-management topics. But the task is built around watching a research talk and interpreting a specific allocator paper, which makes the assignment itself much less exam-like. Part (b) on swapping and (c) on paging are more useful than the research-specific details.

Aufgabe	Topic → subtopic	Type of task	Exam relevance	How exam-like?	Assessment
5	Memory manage- ment → internal vs. external fragmenta- tion; logical vs. physical addresses	Conceptual defini- tion/comparison	High	High	Very good exam-prep question. These are fundamental distinctions explicitly covered in the summary. ibn- zusammenfassung Short “explain the difference between X and Y” questions are also exactly the sort of thing that can appear as exam subquestions.

Aufgabe	Topic → subtopic	Type of task	Exam relevance	How exam-like?	Assessment
6	Memory manage- ment → contiguous allocation; First Fit, Best Fit, Worst Fit	Calculation / algorithm simulation	Medium– High	High	Very exam-like format: small concrete data set, apply three algorithms by hand. The summary explicitly lists First Fit, Best Fit and Worst Fit as the allocation strategies for holes. ibn- zusammenfassung It has not appeared explicitly in the two exam documents we have, so I would put it below paging, but it is exactly the kind of calculation that could easily be asked.

Aufgabe	Topic → subtopic	Type of task	Exam relevance	How exam-like?	Assessment
7	Paging → splitting a logical address into page number + offset	Calculation + tiny coding component	Very High	Very High	One of the strongest exam-prep tasks on the sheet. The summary explicitly requires un- derstanding page number and offset. ibn- zusammenfassung More importantly, the 2016 exam asks essentially this exact operation: given a logical address and page size, determine offset and page index. ibn-klausur- 2016 The recent memory protocol again says logi- cal/physical addresses and page calculations appeared. Gedächtnis- pro- tokoll_Klausur_ IB...

The actual sheet confirms this progression very clearly: Aufgaben 1–3 are named pipes, Win32 handles/redirection, and anonymous-pipe communication; Aufgaben 4–7 then move to fragmentation/swapping/paging, memory-management terminology, placement algorithms and page-number/offset calculations. ueb04-ibn-SS2025 ueb04-ibn-SS2025

Topics

Topics: Interprocess Communication (IPC), Pipes, I/O and file-handle abstractions, Memory Management, Swapping, Fragmentation, Contiguous Memory Allocation, Logical vs. Physical Addressing, Paging

If we compress that to the **main syllabus-level topics**, rather than listing every subtopic:

Topics: Interprocess Communication, Memory Management, Paging

For exam preparation, I would mentally divide Blatt 4 into roughly three priority groups: **Aufgabe 7 and 5 are especially important**, Aufgabe 6 is also worth being able to solve comfortably, while Aufgaben 1 and 3 are useful IPC knowledge but less strongly represented in the exam evidence. Aufgabe 4 contains important theory inside a relatively non-exam-like research exercise, and Aufgabe 2 is the clearest candidate to deprioritize.

1.5 Sheet 5

Yes. **Übungsblatt 5 is unusually exam-relevant overall.** Unlike some earlier sheets, almost the entire sheet is concentrated on one major syllabus block—**Paging and page tables**—and that block appears prominently in both exam documents. In the more recent *Gedächtnisprotokoll*, Paging/Memory Management was even described as a comparatively extensive exam problem, involving construction of page tables, logical/physical addresses, unit conversions, and an inverted-page-table question. *Gedächtnisprotokoll_Klausur_IB...* The 2016 exam likewise devoted an entire 12-point problem to page-table sizes, inverted page tables, address decomposition, and page-table contents. ibn-klausur-2016

Main syllabus-level topics

The sheet belongs almost entirely to **Speicherverwaltung → Paging und Seitentabellen**. More specifically, I would place its material under:

- **Paging and address translation**
 - logical vs. physical addresses
 - page number, frame number, offset
 - direct/normal page tables

- **Page-table organizations**
 - direct page tables
 - inverted page tables
 - hierarchical / multilevel page tables
 - memory consumption of page tables
- **Translation Lookaside Buffer (TLB)**
 - TLB hit / miss
 - TLB capacity
 - locality and repeated accesses
 - TLB invalidation on context switches
 - effective memory-access time
- **Hardware/software interaction in virtual memory**
 - process-specific address mappings
 - why address translation state changes on a process switch
- **Practical address translation**
 - following a translation through a simulator
 - decomposing virtual addresses across levels of a page table

That matches the organization of the summary very closely: the summary's section **"7. Paging und Seitentabellen"** starts with pages, frames, address decomposition and page tables, and then covers TLBs, effective access time, direct/inverted tables and multilevel page tables.
 ibn-zusammenfassung ibn-zusammenfassung ibn-zusammenfassung

Analysis of Übungsblatt 5

Aufgabe	Topic / subtopic	Type of task	Exam relevance	How exam-like?	Assessment
1	Paging; direct page table; mapping page number → frame number ; sparsity/size of a direct table	Conceptual analogy + estimation	High	Medium	The student- ID/address analogy itself is pedagogical and somewhat assignment- like, but the underlying ideas—what indexes a page table, what its entries contain, and why a direct table can be huge/sparse— are core exam material. ueb05-ibn- SS2025

Aufgabe	Topic / subtopic	Type of task	Exam relevance	How exam-like?	Assessment
2	Address translation; page number, offset, frame number; constructing direct page-table entries	Calculation + table construction	Very high	Very high	One of the most important tasks on the sheet. Given logical→physical address mappings, you reconstruct the relevant page-table entries. This is extremely close to actual exam material. ueb05-ibn- SS2025
3	TLB; page size; stride accesses; capacity misses; tem- poral/spatial locality	Calculation / reasoning from code	Medium– High	Medium	Important conceptual TLB exercise, but more elaborate than what we have direct evidence for in the exams. Excellent for understand- ing <i>why</i> a TLB works, rather than a direct historical exam clone. ueb05-ibn- SS2025

Aufgabe	Topic / subtopic	Type of task	Exam relevance	How exam-like?	Assessment
4	TLB invalidation; process switch vs. thread switch; process- specific page tables	Conceptual explanation + coun- terexample	High	Medium– High	Good exam-style conceptual question: explain a mechanism and construct a failure scenario. Not directly repeated in the supplied exams, but squarely within core paging/TLB knowledge. ueb05-ibn- SS2025
5	Multilevel translation + TLB lookup	Simulator / practical supplemen- tary exercise	Low– Medium	Low	The concepts are important, but “run simulator and submit screenshot” is obviously not an exam format. Use it for visualization, not as a priority exam exercise. ueb05-ibn- SS2025

Aufgabe	Topic / subtopic	Type of task	Exam relevance	How exam-like?	Assessment
6	Direct page-table memory size; inverted page table; RAM overhead	Calculation	Extremely high	Extremely high	Probably the strongest exam- preparation task on the sheet. The 2016 exam asks essentially these calculations, and the older minitest included in the exam document contains almost the <i>same</i> <i>wording</i> . ueb05-ibn- SS2025

Aufgabe	Topic / subtopic	Type of task	Exam relevance	How exam-like?	Assessment
7	TLB hit ratio; hierarchical page tables; expected access time; locality	Formula derivation + conceptual explanation	High	High	Very exam- suitable: derive an expected- time formula, then explain why the simplistic probabilistic model is unrealistic. Combines calculation with conceptual understand- ing. ueb05-ibn- SS2025
8	Multilevel page tables; splitting address bits; entries per level; table contents; maximum memory use	Diagram + calculation + conceptual explanation	High	Medium– High	Very useful and compre- hensive, but somewhat longer than a typical short exam subquestion. The individual skills it tests are highly examinable. ueb05-ibn- SS2025

Why Aufgabe 2 and Aufgabe 6 stand out

Aufgabe 2 is almost exactly the skill tested in the recent exam protocol. The protocol says that the Paging problem required students to **create/fill page tables** and

determine **logical and physical addresses**, with PIDs and addresses given. Gedächtnisprotokoll_Klausur_IB... That is essentially what Aufgabe 2 trains.

The 2016 exam is equally direct. It gives a logical address translated to a physical address, asks for its **offset, page index and frame number**, and then asks for the corresponding entries of both a normal and an inverted page table. ibn-klausur-2016

So for exam preparation I would treat **Aufgabe 2 as mandatory**.

Aufgabe 6 is even more striking. The 2016 exam asks for the memory requirement of a normal page table and then for the memory requirement of an inverted page table. ibn-klausur-2016 Moreover, the old 2012 minitest included in the same PDF asks:

- maximum memory requirement of a page table without hierarchical tables;
- what fraction of physical memory an inverted page table would require. ibn-klausur-2016

That is essentially **Übung 5, Aufgabe 6**, just with the parameters packaged slightly differently. So this is not merely “the same topic”; it is a historically demonstrated **exam question form**.

Aufgabe 1: important concept, less important packaging

The point of Aufgabe 1 is to make you understand the structural relationship

$$\text{page number} \longrightarrow \boxed{\text{page table}} \longrightarrow \text{frame number}.$$

That is exactly the formulation used in the summary: the **page number is the index**, and the corresponding page-table entry contains the **frame number**. ibn-zusammenfassung

The student-residence analogy itself is unlikely to appear verbatim on an exam. But once you strip away the analogy, it tests two important questions:

1. What does a direct page table actually represent?
2. Why can a direct page table contain an enormous number of unused entries?

So I would **understand it thoroughly but not spend much time rehearsing the dormitory analogy itself**.

Aufgabe 3: TLB capacity + locality

This one is more sophisticated. You have a 4 KiB page, 4-byte integers, an array accessed with stride (M), and a TLB containing 64 entries. You have to reason about when successive array accesses hit different pages and when the working set exceeds the TLB capacity. ueb05-ibn-SS2025

The summary explicitly treats the TLB as a cache of page-table entries, including TLB hits/misses and the fact that only a small number of pages tend to be accessed frequently. ibn-zusammenfassung

So the **knowledge is definitely syllabus material**, but the exact “choose (M) and (N)” formulation looks more like a deeper exercise than a historically observed exam problem. I’d therefore study the reasoning, but not prioritize it above Aufgaben 2 and 6.

Aufgabe 4: TLB and context switches

This question tests an important consequence of the fact that **different processes have different page tables**.

The summary explicitly says that each process needs its own direct page table because different processes may use the same logical addresses while referring to different data. ibn-zusammenfassung

Therefore stale entries such as

$$p \rightarrow f_A$$

from process A cannot normally simply remain valid when process B, which may map the same page (p) to (f_B), starts executing.

The “does this also apply to threads?” part is particularly useful because it checks whether you understand that threads of one process normally share the same address space.

This is a good **short conceptual-exam-question style**, even though neither of our two exam documents directly asks it.

Aufgabe 5: useful, but supplementary

This is the easiest one to deprioritize. It literally asks you to use the Address Translation Simulator and submit a screenshot. ueb05-ibn-SS2025

The **translation process it illustrates is important**. The activity itself is not.

For exam preparation I’d classify it:

Supplementary visualization exercise — do once if the mechanism is unclear; otherwise skip during final revision.

Aufgabe 7: very useful calculation pattern

This task combines two things:

expected access time

and

locality / why TLB hit rates are high.

The summary explicitly contains an expected-access-time calculation for a TLB:

$$E[X] = h(1 + T) + (1 - h)(2 + T)$$

for the simple one-level case, together with the interpretation of TLB hit and miss costs.
ibn-zusammenfassung

Aufgabe 7 generalizes essentially the same reasoning to a **(k)-level page table**. ueb05-ibn-SS2025

That makes it quite valuable: it is not merely memorizing a formula—you need to reconstruct the formula from the number of memory accesses.

I would therefore consider it **strong exam preparation**, even though our two exam samples do not show this exact question.

Aufgabe 8: multilevel tables

This is the sheet's most comprehensive page-table-structure problem. You have to divide a 32-bit logical address among:

- page offset,
- level-1 index,
- level-2 index,
- level-3 index,

and then determine the number of table entries and total maximum memory usage. ueb05-ibn-SS2025

The summary explicitly expects understanding of this structure: upper levels contain addresses of lower-level page tables, while only the lowest level plays the role of the ordinary direct page table. ibn-zusammenfassung

So this is **core syllabus material**. Its only downside from an exam-preparation perspective is that the exercise is rather extensive. An actual exam might split it into smaller subquestions.

My exam-preparation grouping

If we use the same practical categories we were using for the previous sheets, I would group them approximately like this:

Priority	Tasks	Reason
Must be able to solve confidently	2, 6	Direct overlap with documented exam questions
Strong exam preparation	4, 7, 8	Core mechanisms + highly exam-suitable reasoning/calculation
Important understanding, somewhat more exercise-like	1, 3	Core material, but the particular formulation is less directly evidenced by exams
Supplementary	5	Simulator activity rather than an exam question

And one especially useful conclusion from this sheet is that **“Paging und Seitentabellen” should be treated as a major exam block rather than a peripheral chapter**. The 2016 exam allocated **12/52 points** to Speicherverwaltung, with almost the entire problem centered on page tables. ibn-klausur-2016 The newer memory protocol again reports a large Paging/Memory-Management problem involving page tables, address translation, size calculations, and inverted page tables. Gedächtnisprotokoll_Klausur_IB...

So compared with Übungsblätter 2–4, **Übungsblatt 5 has a very high concentration of material that maps directly onto demonstrated IBN exam content**.

1.6 Sheet 6

Main syllabus-level topics

1. Memory management
2. Paging and page tables
3. Virtual memory and page replacement
4. Locality and working sets
5. Segmentation and contiguous memory allocation

Detailed subtopics

- Page-reference sequences and page faults
- FIFO, LRU, Clock/Second Chance and OPT
- Comparing page-replacement algorithms
- Hardware-supported approximation of LRU
- Reference/accessed bits (**R-bits**)
- Epoch length and page-table scanning overhead
- Working set $WS(t, \Delta)$
- Computing the largest working set in a reference sequence
- Direct page-table size in IA-32
- Memory utilization and fragmentation
- Heap and stack growth
- Comparison of contiguous allocation, segmentation and paging

There is no real **research task** on this sheet. The main categories are calculations, algorithm simulation, conceptual reasoning, pseudocode and practical programming.

Assignment-by-assignment classification

Task	Topic and subtopic	Type	Exam relevance	Exam-like?	Concise assessment
1a	Virtual memory → FIFO and LRU page replacement	Calculation / manual simulation	Very high	Very high	Almost exactly the kind of page-fault calculation already used in a mini-test: simulate frames and count faults.
1b	Page replacement → Clock versus LRU	Constructive example / algorithm comparison	High	Medium–high	Requires genuine understanding of how the two algorithms choose victims, not merely memorizing definitions.
2	Page replacement → mathematical relation between OPT and LRU	Theoretical/concrete reasoning	Medium	Low–medium	An interesting theoretical insight, but less representative than concrete page-fault calculations.

Task	Topic and subtopic	Type	Exam relevance	Exam-like?	Concise assessment
3	Page replacement → implementation of Clock or LRU	Programming / practical implementation	Medium for understanding	Very low	Writing and submitting a complete program is unrealistic in a written exam. Understanding its state transitions remains useful.
4a	Paging → R-bits, approximate LRU and page-table scanning	Calculation / performance analysis	Very high	High	Closely related to the 2016 question about approximating the last-access time using hardware R-bits.
4b	Virtual memory → choice of epoch length	Estimation / design reasoning	High	Medium	More open-ended than a typical calculation; you must state assumptions and balance accuracy against overhead.

Task	Topic and subtopic	Type	Exam relevance	Exam-like?	Concise assessment
5	Working sets → membership and possible overlap	Short conceptual question	Medium–high	Medium–high	A compact definition-and-justification question. Working-set definitions have appeared in the supplied exam material.
6	Working sets → maximum distinct pages in a sliding window of length ()	Algorithm design / pseudocode	High	Medium	The core sliding-window algorithm is exam-suitable, although the file-processing framing is somewhat assignment-like.

Task	Topic and subtopic	Type	Exam relevance	Exam-like?	Concise assessment
7	Memory management → contiguous allocation vs segmentation vs paging	Conceptual comparison	High	Medium–high	A classic explain-and-compare question. Paging is demonstrably exam-relevant; segmentation is less visible in the supplied exams but belongs to the core syllabus.

Recommended study priority

Priority A — practise thoroughly

- **1a:** FIFO/LRU calculation
- **4a:** R-bit scanning calculation
- **4b:** epoch-length reasoning
- **7:** comparison of memory-management schemes

These combine central syllabus material with realistic written-exam formats.

Priority B — understand and be able to reproduce

- **1b:** Clock versus LRU
- **5:** working-set membership
- **6:** largest-working-set pseudocode

These test whether you understand the algorithms and definitions beyond memorization.

Priority C — understand the main idea

- **2:** OPT–LRU mathematical relationship

Useful theoretical knowledge, but comparatively specialized.

Priority D — optional implementation work

- **3:** full simulator implementation

For exam preparation, manually tracing the simulator’s states matters much more than writing the complete program.

Overall, **sheet 6 has high exam relevance**. The strongest evidence is that page-replacement calculations appear almost directly in the supplied material, the 2016 exam asks about R-bit-based recency approximation, and the recent exam recollection describes paging as a substantial exam topic.

1.7 Sheet 7

Main syllabus-level topics

Assignment sheet 7 is primarily about:

1. **File systems**
 - Links, FAT, I-nodes, directories and pathname resolution
2. **Secondary-storage management**
 - Hard-disk access times and disk-scheduling algorithms
3. **RAID**
 - Reliability, redundancy and read performance
4. **Paging and page tables**
 - Only in Aufgabe 3, as an analogy to I-nodes

This corresponds closely to Chapter 10, “Dateien, Dateisysteme, Festplatten,” of the summary. 01-ibn-zusammenfassung.pdf

A particularly important observation is that the recent exam memory protocol explicitly mentions **both FAT and I-nodes**. 13-Ged-chnisprotokoll_Klausur_-IBN_Ersttermin.pdf

Detailed subtopics

- Hard links versus symbolic links
- FAT and linked file allocation
- Cluster allocation and traversal
- I-node structure
- Direct and indirect block addressing
- Maximum file-size calculations
- Directory traversal and pathname lookup
- Analogy between paging and I-nodes
- Seek time and rotational latency
- Conversion between RPM and rotational latency
- FCFS, SSTF, SCAN, C-SCAN and C-LOOK
- RAID 0 versus RAID 1
- Failure probabilities
- Parallel and load-balanced reads

Analysis of the individual assignments

The following classification is based on the full sheet. 07-ueb07-ibn-SS2025.pdf

Aufgabe	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
1	File systems: hard vs. symbolic links	Conceptual comparison / recall	Medium–high	High	A typical short conceptual exam question. Know differences involving I-nodes, cross-filesystem links, dangling links and deletion.

Aufgabe	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
2	File allocation: FAT, clusters, allocation and traversal	Programming / simulation / practical application	Very high content relevance	Low for the complete task; high for its concepts	FAT appeared in the recent exam protocol. Writing the entire simulator is unlikely in an exam, but tracing a FAT or allocating clusters is highly relevant.
3	I-nodes and paging: structural analogy, multilevel tables	Conceptual transfer / supplementary deepening	Medium–high	Low–medium	Both constituent topics are important, but the analogy itself is somewhat artificial and less representative of a direct exam question.

Aufgabe	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
4	I-nodes: maximum file size with direct and single-indirect pointers	Calculation	Very high	Very high	One of the strongest exam-preparation tasks. Almost the same type of I-node calculation appears in the older mini-test material appended to the 2016 PDF.
5	Directories and I-nodes: pathname resolution and disk operations	Calculation / process tracing	High	High	Tests whether you understand how each pathname component requires directory-block and I-node access. A classic filesystem exam problem.
6	Hard disks: rotational latency, seek time, total access time, RPM	Numerical calculation	Medium-high	High	The historical information is contextual; the underlying calculations are standard and exam-like.

Aufgabe	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
7	Disk scheduling: FCFS, SSTF, SCAN, C-SCAN, C-LOOK	Algorithm execution / calculation	High	Very high	A classic “execute the algorithm and calculate the cost” task. Pay attention to initial direction and whether the head travels to the physical endpoint.
8	RAID 0/1: failure probabilities	Probability calculation + explanation	Very high	Very high	Especially important: essentially this exact RAID 0/1 probability task occurs in the older mini-test material contained in the 2016 PDF.

Aufgabe	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
9	RAID 0/1: comparative read performance	Conceptual reasoning / performance analysis	Medium–high	High	A good short exam question. Requires distinguishing a single sequential read from multiple independent reads that can be distributed across mirrors.

The comparison material comes from the 2016 examination PDF, including the older mini-tests appended to it. 12-ibn-klausur-2016.pdf

Important distinction for Aufgabe 2

I would not treat the entire FAT simulator as something you must reproduce under exam conditions. Split it into two layers:

- **Exam-relevant:** understanding the FAT representation, following a cluster chain, allocating clusters, calculating the number of required clusters and recognizing error cases.
- **Mostly practical:** designing the class, writing complete code, testing it and printing outputs.

Thus, Aufgabe 2 is simultaneously **very important in subject matter** and **not very exam-like in its full implementation format**.

Recommended exam-preparation priority

1. **Aufgabe 4** — direct I-node calculation and closely matched by older exam material
2. **Aufgabe 8** — essentially an exact previous mini-test question

3. **Aufgabe 2, conceptual portion** — FAT explicitly appeared in the recent exam protocol
4. **Aufgabe 5** — fundamental I-node/directory mechanics
5. **Aufgabe 7** — standard algorithm-execution problem
6. **Aufgaben 1 and 9** — compact conceptual questions
7. **Aufgabe 6** — standard calculations, but less directly supported by the supplied exams
8. **Aufgabe 3** — useful for deeper understanding, but the least directly exam-like formulation

Overall, sheet 7 is **highly relevant**. It contains two exceptionally strong past-exam matches—Aufgaben 4 and 8—and directly practises FAT and I-nodes, which the recent memory protocol identifies as actual exam topics.

1.8 Sheet 8

Main syllabus-level topics

1. **Deadlocks and resource allocation**
2. **CPU scheduling**

Sheet 8 is especially important for exam preparation because deadlocks and the Banker algorithm occur directly in both exam references. Scheduling calculations also occur in an older mini-test included in the 2016 exam document.

The classification below is based on the assignment sheet 08-ueb08-ibn-SS2025.pdf, the syllabus structure in the summary 01-ibn-zusammenfassung.pdf, and comparison with the 2016 exam 12-ibn-klausur-2016.pdf and the recent exam memory protocol 13-Gedchtnisprotokoll_Klausur_-IBN_Ersttermin.pdf.

Detailed subtopics

Deadlocks

- Definition of a resource deadlock
- Four necessary Coffman conditions:
 - mutual exclusion

- hold and wait
- no preemption
- circular wait
- Necessary versus sufficient conditions
- Deadlocks caused by incorrect semaphore acquisition
- Prevention through a globally consistent resource-acquisition order
- Resource-allocation graphs
- Multiple instances of a resource type
- Resource vector E and available-resource vector A
- Allocation matrix C and request matrix R
- Banker algorithm
- Safe execution sequences
- Deadlock detection versus deadlock avoidance

CPU scheduling

- CPU-bound versus I/O-bound processes
- CPU bursts and I/O bursts
- FCFS
- Shortest Job First
- Round Robin and the time quantum
- Gantt diagrams
- Turnaround time
- Waiting time
- Average waiting time
- Fairness and starvation
- Priority scheduling

Analysis of the individual assignments

Aufgabe	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
1	Deadlocks: necessary conditions, multiple resource instances, necessary vs. sufficient	Conceptual proof; counterexample construction	High	High	Very close to the 2016 question asking for graphs with cycles both with and without a necessary deadlock. Although marked bonus, the underlying distinction is exam-relevant.
2	Semaphore deadlock; four Coffman conditions; deadlock prevention	Execution trace; conceptual explanation; design/fix	Very high	Very high	Almost a direct exam pattern: show how opposite lock ordering produces deadlock, identify the four conditions, and remove circular wait through consistent lock ordering.

Aufgabe	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
3	Banker algorithm; matrices (E,C,R,A); identifying deadlocked processes	Programming; algorithm implementation; testing	High conceptually	Medium as a full task	Implementing a complete program is unlikely in a written exam. Understanding the algorithm, writing pseudocode and tracing it manually are extremely relevant.
4	Resource-allocation graph; resource vectors and matrices; manual Banker algorithm	Graph construction; data representation; hand calculation	Very high	Very high	The strongest exam-preparation task on the sheet. The recent exam reportedly used essentially this sequence: draw the graph and apply the Banker algorithm.

Aufgabe	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
5	Scheduling: CPU-bound versus I/O-bound processes	Conceptual explanation	Medium	Medium	A plausible short theory question. Know that I/O-bound processes usually have short CPU bursts and should receive CPU time quickly to keep devices and the system responsive.
6	FCFS, SJF and Round Robin; Gantt diagrams; turnaround and waiting times	Calculation; algorithm trace; comparison	High	High	A standard scheduling calculation. An older mini-test in the 2016 document asks for nearly the same SJF/RR execution trace. Practise this carefully.

Aufgabe	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
7	Starvation under scheduling algorithms	Conceptual classification	Medium–High	Medium–High	A very typical one-point conceptual question. Under the usual assumptions, starvation can occur with SJF and priority scheduling , but not with FCFS or Round Robin.

Small issue in Aufgabe 6

Part (b) refers to „*die vier genannten Scheduling-Algorithmen*“, but part (a) names only three:

- FCFS
- Shortest Job First
- Round Robin

This is almost certainly a typo. Unless another algorithm was specified orally, calculate the values for these three.

Why these tasks are strongly exam-relevant

The match with the exam evidence is unusually direct:

- The **2016 exam** contains an entire 11-point section on deadlocks, including:
 - deadlock definition,
 - semaphore code producing a deadlock,
 - resource-allocation graphs,
 - cycles that do or do not imply deadlock,

- the purpose and main idea of the Banker algorithm.
- The **recent exam protocol** explicitly reports:
 - drawing a resource/allocation graph,
 - applying the Banker algorithm,
 - concluding that no deadlock exists.
- An older mini-test included in the 2016 document asks for **SJF and Round-Robin scheduling traces**, closely matching Aufgabe 6.

Recommended preparation priority

1. **Aufgabe 4** — closest match to the recent exam
2. **Aufgabe 2** — direct classical exam pattern
3. **Aufgabe 6** — central scheduling calculation
4. **Aufgabe 1** — important conceptual distinction about multiple instances
5. **Aufgabe 3, algorithmic core** — understand and trace it; full programming is secondary
6. **Aufgabe 7** — easy, compact exam knowledge
7. **Aufgabe 5** — useful short conceptual background

Overall, **sheet 8 is one of the most exam-relevant sheets so far**, mainly because Aufgaben 2 and 4 correspond almost directly to documented exam questions.

1.9 Sheet 9

Sheet 9 is almost entirely about **computer networks**, progressing through the protocol stack:

1. **Fundamentals of packet-switched networks**
2. **Network delay and queueing**
3. **Internet structure, routing paths and topology**
4. **Protocol layers, encapsulation and addressing**
5. **Transport protocols and reliable communication**
6. **Application-layer protocols, especially HTTP**
7. **IPv4 fragmentation**

This classification is based on the exercises in sheet 9, the networking section of the summary, and the two available exam documents. 09-ueb09-ibn-SS2025.pdf 01-ibn-zusammenfassung.pdf

Detailed subtopics

- Packet switching versus circuit switching
- Statistical multiplexing
- Binomial probability for simultaneously active users
- Queueing delay and transmission time
- Traceroute and interpretation of network paths
- ISPs, BelWü, network redundancy and Internet exchange points
- Internet protocol stack
- Protocol data units: message, segment/datagram, IP datagram and frame
- Endpoints and addressing at different layers
- Encapsulation and protocol robustness
- TCP over extremely long-delay or intermittently connected networks
- HTTP cookies and state management
- HTTP requests and responses
- Conditional HTTP GET and caching
- Relationship between HTTP messages and TCP segments
- IPv4 fragmentation, MTU, identification, flags and fragment offsets

Exercise-by-exercise analysis

Exercise	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
1	Packet switching; statistical multiplexing; binomial probability	Calculation / modelling	Medium	Medium	The underlying idea is important, but the large binomial-tail calculation is more specialised than the observed exam questions. Understand the setup and interpretation.
2	Queueing delay; transmission time (L/R)	Formula derivation / calculation	High	High	Short, self-contained and closely related to earlier exam-style calculations involving transmission and propagation delays.

Exercise	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
3	Traceroute; ISP topology; redundancy; Internet exchange points	Practical investigation / research	Medium	Low	Useful networking intuition, but dependent on live infrastructure and external research. It does not directly practise Dijkstra or subnetting, despite being routing-related.
4	Protocol stack; encapsulation; PDUs; endpoints; addressing	Conceptual knowledge / structured table	Very high	Very high	One of the most important exercises. The 2016 exam contained an extremely similar task about encapsulation, layer headers and address-like information.

Exercise	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
5	Encapsulation; layer-dependent attack scope; protocol security	Conceptual reasoning / security enrichment	Low–medium	Low	Tests whether you understand the scope of transport, network and link-layer communication, but the recursive ZIP scenario is unusual and supplementary.
6	TCP reliability under enormous delay and disrupted connectivity; delay-tolerant networking	Research / conceptual transfer / bonus	Medium	Low–medium	TCP reliability and connection establishment are relevant, but the Mars/DTN setting and required reading make this primarily enrichment.

Exercise	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
7	HTTP cookies; maintaining state over stateless HTTP	Short conceptual explanation	High	High	Directly represented in the summary and well suited to a short written exam question. Know the first response, cookie storage, later request and server-side association.
8	HTTP in Wireshark; GET/response; conditional GET; TCP segmentation; embedded objects	Practical laboratory / protocol analysis	Medium–high	Low as a complete exercise	Wireshark operation is unlikely in a paper exam, but the protocol observations it reveals can become conceptual exam questions.

Exercise	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
9	IPv4 fragmentation; MTU; header length, ID, MF flag and fragment offset	Calculation / header analysis	High	Very high	A classic written-exam calculation. It directly matches the fragmentation material in the summary and requires precise handling of byte lengths and offsets.

The particularly strong rating for Aufgabe 4 comes from the 2016 exam, whose Aufgabe 4 asks for the encapsulated frame, the layer headers and their addressing information. 12-ibn-klausur-2016.pdf The more recent memory protocol emphasizes Dijkstra, forwarding tables and subnetting; this makes Aufgabe 3 only indirectly useful for that routing component. 13-Ged-chnisprotokoll_Klausur_-IBN_Ersttermin.pdf

Recommended exam-preparation priority

First priority

1. **Aufgabe 4** — protocol layers, encapsulation and addressing
2. **Aufgabe 9** — IPv4 fragmentation

These are the most conventionally exam-like tasks on the sheet.

Second priority

3. **Aufgabe 2** — queueing-delay derivation
4. **Aufgabe 7** — HTTP cookies
5. **Aufgabe 1** — statistical multiplexing, especially the general probabilistic model

Selective preparation

6. **Aufgabe 8** — learn the HTTP concepts and packet observations; reproducing the complete Wireshark lab is less important for the written exam.
7. **Aufgabe 6** — learn what very long RTTs and interruptions do to TCP, but the article-specific details are secondary.
8. **Aufgabe 3** — useful practical background, but not a substitute for practising Dijkstra, forwarding tables and subnetting.
9. **Aufgabe 5** — understand the layer-dependent scope of the attack; the specific construction is low priority.

Overall, sheet 9 contains several highly relevant networking fundamentals, but it mixes them with substantial research and laboratory work. For efficient exam preparation, **Aufgaben 4 and 9 deserve the most careful complete solutions.**

1.10 Sheet 10

Yes. Sheet 10 is one of the most exam-relevant sheets so far because it concentrates on the three topics explicitly reported in the newer exam: **subnetting, forwarding tables, and Dijkstra**. 10-ueb10-ibn-SS2025.pdf

Main syllabus-level topics

1. **Network layer and IPv4 addressing**
 - Classful and classless addressing
 - Subnetting and address allocation
 - Network and host portions of an IP address
2. **Packet forwarding**
 - Forwarding/routing tables
 - Prefix representation
 - Longest-prefix matching
3. **Routing algorithms**
 - Link-state routing and Dijkstra's algorithm
 - Distance-vector routing and the Bellman–Ford equation
 - Properties of shortest paths

These correspond closely to the IPv4, subnetting, forwarding, Dijkstra, and Bellman–Ford sections of the course summary. 01-ibn-zusammenfassung.pdf

Detailed subtopics

- IPv4 address classes A, B and C
- CIDR notation such as /16, /23, /25
- Network ID, subnet ID and host ID
- Subnet masks in binary and decimal notation
- Calculating the number of addresses and hosts
- Fixed-size and variable-size subnet allocation
- Dividing address blocks among networks and router links
- Translating address ranges into prefixes
- Prefix ordering and longest-prefix matching
- Constructing forwarding tables
- Traceroute and changing network routes
- Dijkstra tables: (N'), tentative distances and predecessors
- Optimal-substructure property of shortest paths
- Bellman–Ford recurrence
- Why Dijkstra requires nonnegative edge weights

Classification of the individual exercises

Exercise	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
1	IPv4 classes; Whois; traceroute; real network topology	Conceptual + research + practical investigation	Low–medium	Low	Identifying address classes is relevant, but RIPE lookup and performing traceroute are mainly supplementary practical work.

Exercise	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
2	Subnet masks; address counts; network/host ID; subdivision of an address block	Calculation + subnet design	Very high	Very high	Direct practice for dividing networks into subnets and determining prefixes or masks.
3a	CIDR/VLSM address allocation for host networks and router links	Advanced calculation + network design	Very high	High–very high	Covers subnet allocation under capacity constraints. Slightly more intricate than a typical short exam question.
3b	Forwarding tables; route aggregation; longest-prefix matching	Table construction + application	Very high	Very high	Almost exactly the combination of subnetting and forwarding-table construction reported in the newer exam.

Exercise	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
4	Converting address ranges to prefixes; prefix order; longest-prefix matching	Binary calculation + routing-table reasoning	Very high	Very high	Excellent preparation for prefix manipulation and routing decisions, especially overlapping prefixes.
5	Dijkstra's shortest-path algorithm	Algorithm execution + table calculation	Very high	Extremely high	Directly matches both reference exams: execute Dijkstra and fill in the algorithm's table.
6a	Optimal substructure of shortest paths	Proof + theory	Medium	Medium	Useful theoretical justification, but less likely than executing the algorithm.
6b	Traceroute and changing routes	Conceptual application	Low–medium	Low–medium	Tests understanding that routes and routing information can change over time.

Exercise	Topic and subtopic	Type	Exam relevance	Exam-likeness	Concise assessment
6c	Bellman–Ford equation and shortest-path decomposition	Conceptual theory	High	Medium–high	Bellman–Ford appeared in the 2016 exam, although there it was used for a concrete distance-vector update.
6d	Negative edge weights and limitations of Dijkstra	Algorithmic reasoning	Medium	Medium	Important for understanding Dijkstra, but less directly reflected in the reference exams.

Evidence from the reference exams

The newer memory protocol explicitly lists, in a single routing exercise:

- performing Dijkstra,
- filling a routing/forwarding table,
- dividing a network into subnets,
- determining prefixes and subnet masks.

That makes Exercises **2–5** unusually strong direct preparation. 13-Ged-chnisprotokoll_Klausur__-IBN_Ersttermin.pdf

The 2016 exam independently confirms the importance of:

- executing Dijkstra in tabular form,
- deriving forwarding entries from predecessor pointers,
- explaining and applying the Bellman–Ford equation,
- using a subnet mask to determine whether two hosts belong to the same subnet.

Thus Exercise 5 is almost an exact match, while Exercises 2–4 and 6c cover closely related examinable skills. 12-ibn-klausur-2016.pdf

Recommended study priority

1. **Exercise 5** — exact Dijkstra exam practice
2. **Exercise 3** — strongest combined subnetting and forwarding-table exercise
3. **Exercise 2** — essential subnetting calculations
4. **Exercise 4** — prefixes and longest-prefix matching
5. **Exercise 6c** — Bellman–Ford connection
6. **Exercises 6a and 6d** — conceptual reinforcement
7. **Exercise 1 and 6b** — supplementary practical context

Overall, **Exercises 2–5 form the exam core of Sheet 10**. If time is limited, those should all be completed carefully rather than merely read.

1.11 Overview

The third column is useful, especially if it contains only the exercises with the strongest combination of **topic relevance** and **exam-like format**.

Here, “A” means *Aufgabe*. The ratings are absolute, not merely relative to the other exercises on the same sheet.

Sheet	Main syllabus topics	Most exam-like and exam-relevant exercises
1 02-ueb01-ibn-SS2025.pdf	OS foundations; CPU registers and powers of two; protection and privilege modes; interrupts and system calls; kernel stack; POSIX/Unix interfaces; shell scripting	A2(ii) — powers-of-two calculations (medium–high); A3–4 — protection, kernel mode and kernel stack (medium)

Sheet	Main syllabus topics	Most exam-like and exam-relevant exercises
2 03-ueb02-ibn-SS2025.pdf	Processes and threads; PCB; <code>fork()</code> and process hierarchies; process states; <code>exec()</code> ; context switches; pthreads	A2(a) — tracing <code>fork()</code> and drawing the process tree (very high); A3 — process hierarchy and process states (very high)
3 11-ueb03-ibn-SS2025.pdf	Process synchronization; race conditions; critical sections; semaphores, mutexes and condition variables; producer-consumer; readers-writers	A1 — detect and prevent a race condition (very high); A2 — synchronization primitives (high); A4(a–b) — trace and correct semaphore-based synchronization (high)
4 04-ueb04-ibn-SS2025.pdf	Interprocess communication and pipes; I/O redirection; memory management; fragmentation and swapping; contiguous allocation; logical and physical addresses; paging basics	A5 — core memory-management distinctions (high); A6 — First/Best/Worst Fit calculation (medium–high); A7 — page number and offset (high)
5 05-ueb05-ibn-SS2025.pdf	Paging and address translation; page tables; hierarchical and inverted page tables; TLB; address-space calculations; effective access time	A2 — construct page-table entries from address mappings (very high); A6 — direct and inverted page-table size (very high); A8 — multilevel page tables (very high)
6 06-ueb06-ibn-SS2025.pdf	Virtual memory; page replacement; FIFO, LRU, Clock and OPT; reference bits; working sets; segmentation versus paging	A1 — execute FIFO/LRU/Clock-style replacement (very high); A4 — reference bits and LRU approximation (high); A5 — working-set concept (medium)

	Sheet	Main syllabus topics	Most exam-like and exam-relevant exercises
7	07-ueb07-ibn-SS2025.pdf	File systems; hard and symbolic links; FAT; i-nodes; directory traversal; disk performance and scheduling; RAID	A4 — maximum file size from direct/indirect i-node pointers (very high); A5 — directory traversal and disk operations (very high); A2 — FAT structure (high relevance , but implementation format is less exam-like)
8	08-ueb08-ibn-SS2025.pdf	Deadlocks; necessary deadlock conditions; resource-allocation graphs; Banker's algorithm; CPU scheduling; starvation	A2 — semaphore deadlock, conditions and prevention (very high); A4 — resource graph, matrices and Banker's algorithm by hand (extremely high); A3 — Banker's algorithm (high relevance , but programming format is less exam-like)
9	09-ueb09-ibn-SS2025.pdf	Network performance and delays; packet switching; protocol layers; encapsulation and addressing; TCP; HTTP and cookies; IPv4 fragmentation	A4 — layers, encapsulation, endpoints and addressing (extremely high); A9 — IPv4 fragmentation calculation (high); A2 — queueing-delay calculation (medium-high)
10	10-ueb10-ibn-SS2025.pdf	IPv4 addressing and subnetting; CIDR and prefixes; forwarding tables; longest-prefix matching; Dijkstra; Bellman-Ford and distance-vector routing	A2 — subnet masks and network/host IDs (very high); A3 — subnet allocation and forwarding tables (extremely high); A4 — prefixes and longest-prefix matching (very high); A5 — Dijkstra table (extremely high)

The syllabus classification follows the organization and terminology of the course summary. 01-ibn-zusammenfassung.pdf

The “exam-like” judgments are based particularly on the direct matches in the 2016 exam—race conditions and semaphores, page tables, deadlocks, protocol layers, Dijkstra and Bellman–Ford—and in the newer memory protocol—`fork()`, race conditions, Banker’s algorithm, FAT/i-nodes, paging, subnetting, forwarding tables and Dijkstra. 12-ibn-klausur-2016.pdf 13-Ged-chnisprotokoll_Klausur_-IBN_Ersttermin.pdf

Strongest sheets overall

If you prioritize entire sheets rather than individual exercises:

1. **Sheet 10** — subnetting, forwarding tables and Dijkstra
2. **Sheet 8** — deadlocks, resource graphs and Banker’s algorithm
3. **Sheet 5** — paging and page tables
4. **Sheet 3** — race conditions and synchronization
5. **Sheet 2** — `fork()` and process states
6. **Sheet 7** — FAT and i-nodes
7. **Sheet 9** — network layers and addressing

Sheets **1**, **4** and **6** still contain useful exam material, but their exercises have fewer direct matches with the two reference exams.

2 Plan

Yes—this is still realistic, especially because you solved the material before. You are **reactivating knowledge**, not learning the entire course from zero.

The crucial change is: **do not redo all ten sheets sequentially**. They contain research, programming, and practical tasks that are useful pedagogically but inefficient for a written exam. The sheets collapse into about seven examinable clusters.

2.1 Priority hierarchy

Priority 1: direct reference-exam matches

These should become automatic:

- `fork()`, process trees and process states
- Race conditions, semaphores and synchronization
- Deadlocks, resource-allocation graphs and Banker's algorithm
- Paging, page-table calculations and address translation
- FAT and i-nodes
- Protocol layers, encapsulation and addressing
- Subnetting, prefixes and forwarding tables
- Dijkstra and Bellman–Ford

Priority 2: likely supporting material

- Page replacement: FIFO, LRU and Clock
- TLB and reference bits
- CPU scheduling
- IP fragmentation
- Disk scheduling and RAID
- Memory fragmentation and allocation algorithms

Priority 3: postpone unless time remains

- Long programming exercises
- Internet research and API-detail questions
- Traceroute, Whois and Wireshark experiments
- Historical hardware details
- Learning-methodology questions
- Reimplementing FAT, Banker, LRU or producer–consumer in code

For programming exercises, understand the underlying algorithm and execute it **by hand**. Do not spend hours coding it again.

2.2 Ten-day core plan

Assume approximately 4–6 focused hours per day. If you have 11–13 days, use the extra days for consolidation rather than expanding the syllabus.

Day	Focus	Essential exercises and outcomes
1	Diagnostic + processes	Attempt the 2016 exam under time pressure, even if incomplete. Then review Sheet 2 A2(a), A3. Know <code>fork()</code> return values, process trees and the five process states.
2	Synchronization	Sheet 3 A1, A2 and A4(a–b). Trace a race condition instruction by instruction; repair it with semaphores; understand <code>wait</code> , <code>signal</code> , mutexes and condition variables.
3	Paging fundamentals	Sheet 4 A5, A7; Sheet 5 A2, A6. Split addresses into page number/offset, construct page-table entries, and calculate normal/inverted page-table sizes.

Day	Focus	Essential exercises and outcomes
4	Advanced paging and virtual memory	Sheet 5 A8; Sheet 6 A1, A4. Multilevel tables, TLB, FIFO/LRU/Clock and reference-bit approximation.
5	Deadlocks and scheduling	Sheet 8 A2 and A4 first; A6 if time remains. Know the four deadlock conditions, resource graphs, Banker matrices and hand execution.
6	File systems	Sheet 7 A4 and A5; review the FAT mechanism from A2 without implementing it. Practise i-node maximum-file-size and pathname traversal calculations.
7	Network stack	Sheet 9 A4 and A9. Reconstruct encapsulation, endpoints and addressing at each layer; then practise IP fragmentation.
8	Subnetting and forwarding	Sheet 10 A2, A3 and A4. Work entirely from scratch: masks, address counts, network/host IDs, prefixes and longest-prefix matching.
9	Routing algorithms	Sheet 10 A5 and A6(c); 2016 exam Question 5. Complete Dijkstra tables, derive next hops from predecessor pointers, and perform Bellman–Ford updates.

	Day	Focus	Essential exercises and outcomes
	10	Timed mixed mock + repair	Assemble unused exercise parts into a 90-minute mock. Mark it immediately. Redo every incorrect question from a blank page without consulting the solution.

2.3 If you have 11–13 days

Use the additional days as follows:

- **Day 11:** Repair your weakest two topics.
- **Day 12:** Second full timed mock covering all major clusters.
- **Day 13:** Light closed-book recall, formulas and common mistakes. No major new topics.

If you have only ten days, the core plan already covers the important material. Do not sacrifice mock-exam practice to squeeze in every supplementary topic.

2.4 How to structure each study day

A good five-hour day would look like this:

1. 30 minutes: closed-book retrieval

- Write everything remembered from yesterday.
- Reproduce algorithms, definitions and calculation procedures.

2. 60–75 minutes: targeted theory recovery

- Use the summary only to repair what you could not recall.
- Avoid reading an entire chapter passively.

3. Two 75-minute problem blocks

- Solve selected exercises without notes.
- Work in a handwritten, exam-style format.
- Impose time limits.

4. 30 minutes: correction and error log

- Record each substantive mistake.
- Classify its cause: forgotten fact, wrong method, arithmetic error or misreading.

5. 20–30 minutes: delayed repetition

- Reattempt one problem from two or three days earlier.

If you have only three hours, keep the same structure but shorten the theory section—not the problem-solving section.

2.5 Build an error log, not another long summary

Use five columns:

Topic	Mistake	Cause	Correct rule/method	Retest
Subnetting	Used 2^h as usable-host count	Forgot net-work/broadcast addresses	Normally $2^h - 2$, unless stated otherwise	Tomorrow
Dijkstra	Updated an already-finalized node	Algorithm rule forgotten	Nodes in N' are final	In two days

Your final revision should be driven primarily by this list.

2.6 Learn reusable solution templates

For each major exam type, have a standard procedure:

- **fork():** draw one branch per call; label parent and child return values.
- **Race condition:** expand operations into load–modify–store and show the interleaving.
- **Banker:** calculate available resources, find a satisfiable process, release its allocation, repeat.
- **Paging:** derive offset bits from page size, then split the logical address.
- **I-node:** calculate pointers per indirect block before calculating capacity.
- **Subnetting:** determine required host bits, then prefix length, block size and aligned boundaries.
- **Longest-prefix matching:** list all matching prefixes and choose the longest.
- **Dijkstra:** maintain N' , $D(v)$ and $p(v)$ systematically.
- **Bellman–Ford:** minimize link cost plus the neighbour’s advertised distance.

2.7 The most important rule

Do not measure progress by “pages read” or “sheets reviewed.” Measure it by:

Can I solve this problem from a blank page, correctly and within exam time?

After about five days, some topics will already feel substantially more familiar. That is the advantage of having solved the sheets before. A focused ten-day recovery is much more plausible than the current breadth of the material makes it appear.

3 Processes and fork()

3.1 Unit 1: fork(), process trees, and process states

This is a particularly high-priority unit: the newer exam contained code using `fork()` and asked for the number of processes, return values, executed branches, and the five process states. 13-Ged-chnisprotokoll_Klausur_-IBN_Ersttermin.pdf

1. What is a process?

A **program** is passive executable code stored in a file. A **process** is an instance of a program currently being executed.

A process includes:

- its program code;
- stack, heap and global data;
- CPU state, such as the program counter and registers;
- operating-system information stored in its **Process Control Block (PCB)**:
 - process ID;
 - current process state;
 - scheduling information;
 - memory-management information;
 - open resources.

Multiple processes can therefore execute the same program while having different variables, stacks, registers and process IDs.

3.2 2. What does `fork()` do?

In a POSIX system,

```
pid_t p = fork();
```

creates **one new child process**.

Immediately after a successful call, there are two processes:

- the original **parent process**;
- the newly created **child process**.

Both continue execution at the instruction immediately following `fork()`.

Conceptually, the child receives a copy of the parent's state at the moment of the call:

- program code;
- stack, heap and global variables;
- environment;
- open file descriptors;
- current position in the program.

Ordinary variables are subsequently independent. Changing a variable in the child does not change the parent's corresponding variable.

In practice, the memory pages are usually initially shared using **copy-on-write** and copied only when one process modifies them. For the exam, however, you can reason as if each process has its own copy.

Return value

The processes distinguish themselves using the return value:

Result of <code>fork()</code>	Meaning
<code>p > 0</code>	This is the parent; <code>p</code> is the child's PID
<code>p == 0</code>	This is the child
<code>p == -1</code>	Creation failed; no child was created

A very common pattern is therefore:

```
pid_t p = fork();

if (p == 0) {
    // Executed by the child
} else if (p > 0) {
    // Executed by the parent
} else {
    // Error handling
}
```

The exact positive PID is normally unknown unless it is provided.

Crucial distinction

`fork()` is called once, but it **returns twice**:

- once in the parent, with a positive value;
- once in the child, with zero.

3.3 3. Counting processes

Every process that reaches a successful `fork()` creates one additional process.

Thus, if there are (k) processes immediately before a `fork()` and all (k) execute it successfully, there will be

$$2k$$

processes afterwards.

For n successive unconditional calls,

```
fork();
fork();
fork();
```

the total number is

$$2^n.$$

Here, three calls produce $2^3 = 8$ processes, including the original process.

This shortcut works only when **every existing process executes every call**. With conditions, loops, `exit()`, or `exec()`, you must trace the individual execution paths.

3.4 4. The safe process-tree method

For each `fork()`:

1. Identify every process that reaches the call.
2. Give each of them one new child.
3. Copy the parent's current variable values into the child.
4. Assign the appropriate return value:
 - parent receives a positive PID;
 - child receives zero.
5. Continue tracing every resulting process separately.

Do not count the number of occurrences of the word `fork` and immediately use 2^n . A conditional call may be reached by only part of the process tree.

Example

```
pid_t p = fork();  
  
if (p == 0)  
    fork();
```

After the first call:

- parent: `p > 0`;
- child: `p == 0`.

Only the child enters the `if` and executes the second call. Consequently, there are three processes, not four.

3.5 5. Process hierarchy

Each child has:

- its own PID;
- the PID of its parent, called its PPID.

This produces a process tree. A child created later by an existing child becomes a **grandchild** of the original process.

The parent and child normally execute concurrently. The scheduler may run either one first, so the order of their output is generally not predictable.

However, the order of instructions **inside one process** remains fixed.

3.6 6. The five basic process states

The five-state model required by the course is: 01-ibn-zusammenfassung.pdf

State	Meaning
New	The process is being created.
Ready	It can execute but is waiting for a CPU.
Running	Its instructions are currently being executed by a CPU.
Waiting / Blocked	It cannot continue until some event occurs, such as completion of I/O.
Terminated	It has finished execution or has been terminated.

The important distinction is:

- **Ready:** capable of running, but no CPU currently assigned;
- **Waiting:** not capable of continuing yet, even if a CPU were available.

State transitions

Transition	Typical cause
New → Ready	The process is admitted into the ready queue
Ready → Running	The scheduler dispatches it
Running → Ready	Time slice expires or the process is preempted
Running → Waiting	It requests I/O or waits for an event
Waiting → Ready	I/O completes or the awaited event occurs
Running → Terminated	It finishes or calls <code>exit()</code>

Normally, a process does not move directly from Waiting to Running. Once its event occurs, it becomes Ready and waits to be selected by the scheduler.

Where does `fork()` fit?

A parent must be **Running** to execute `fork()`. The operating system creates a child, which passes through New and becomes Ready. The scheduler then decides whether the parent or the child runs next.

Calling `fork()` does **not** guarantee that the child executes first.

3.7 7. Common exam traps

- Counting only the children and forgetting the original parent.
- Assuming that only the child continues after `fork()`.
- Reversing the return values: the child gets 0; the parent gets the child's PID.
- Assuming the parent always runs before the child.
- Forgetting that descendants inherit the current value of previously assigned variables.
- Using (2^n) even though some calls are conditional.
- Confusing Ready with Waiting.
- Claiming that I/O completion causes Waiting → Running directly.

Sheet 2 contains the corresponding course practice involving return values, conditional calls and process trees. 03-ueb02-ibn-SS2025.pdf

4 Exam-like exercise

Try this closed-book and give yourself approximately **15 minutes**.

Assume that every `fork()` succeeds.

```
#include <stdio.h>
#include <unistd.h>

int main(void)
{
    int x = 0;

    pid_t p = fork();           // F1

    if (p == 0) {
        x = 1;
        fork();                 // F2
    } else {
        x = 2;
        fork();                 // F3
    }

    if (p == 0) {
        x = x + 10;
        fork();                 // F4
    }

    printf("p-category=%s, x=%d\n",
           p == 0 ? "zero" : "positive", x);

    return 0;
}
```

Part A: `fork()` and the process tree

1. How many processes exist at the end, including the original process?

2. How many times is `printf()` executed?
3. Draw the process tree. Label the original process (`P_0`), and label each new process according to the call that created it.
4. How many final processes have:
 - `p == 0`;
 - `p > 0`?
5. Which processes execute:
 - the first `if` branch;
 - the `else` branch;
 - the second `if` branch?
6. What is the final value of `x` in each process?
7. Is the order of the printed lines uniquely determined? Explain briefly.
8. Can the exact positive value of `p` be determined from the code alone? What does that value represent?

A table may make your tracing easier:

Final process	Parent	Created by	<code>p</code> category	Final <code>x</code>	Executes second <code>if</code> ?
P_0	—	—			
...					

Part B: Process states

Give the state transition caused by each event:

1. A process has just been admitted by the operating system and may now compete for the CPU.
2. The scheduler selects it for execution.
3. Its time slice expires.
4. It is selected again and then requests data from the disk.
5. The disk operation finishes.
6. The scheduler selects it again.
7. The process calls `exit()`.

Write the answer in the form:

Ready → Running

Send me your solution—even if parts are uncertain—and I'll correct it step by step before we move to race conditions and synchronization.

4.1 Answers

Part A

1. there are 6 processes
2. each process executes printf once, therefore 6 times

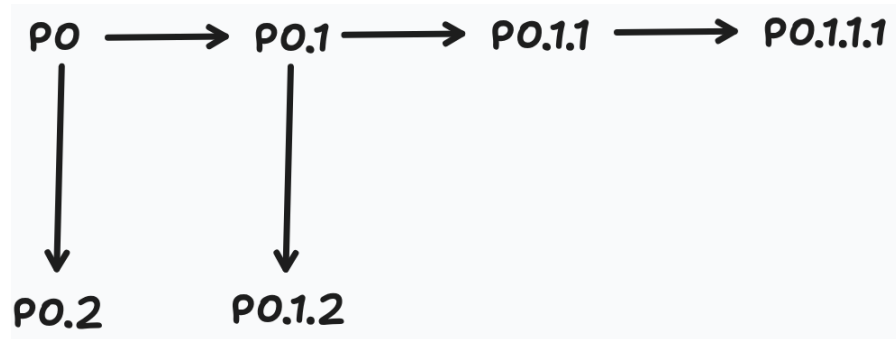


Figure 4.1: process-tree

- 3.
4. P0, P0.2 have $p > 0$, i.e. 2 processes have positive p , and P0.1, P0.1.1, P0.1.1.1, P0.1.2 have $p == 0$, i.e. 4 have zero p .
5. first if branch executed by: P0.1
else branch: P0
second if branch: P0.1, P0.1.1
6. final x value of each process: P0: 2
P0.1: 11
P0.1.1: 11
P0.1.1.1: 11
P0.2 : 2
P0.1.2: 11
7. no, because the scheduler arbitrarily decides in which order the parent and children processes are scheduled
8. No, a positive p value represents the PID of the generated child process and is internally determined by the OS

Part B

1. new -> ready
2. ready -> running
3. running -> ready
4. running -> waiting
5. waiting -> ready
6. ready -> running
7. running -> terminated

5 Race conditions, semaphores and synchronization

This is one of the highest-priority units. The 2016 exam devoted an entire 11-point problem to instruction-level race tracing, semaphore repair, semaphore semantics and execution ordering. 12-ibn-klausur-2016.pdf The recent exam protocol also reports an independent pseudocode-based race-condition problem. 13-Ged-chnisprotokoll_Klausur_-IBN_Ersttermin.pdf

The coverage below follows Day 2 of your plan and the selected Sheet 3 exercises. 15-Operating-Systems-and-Networks-SoSe-25-Notes.pdf 11-ueb03-ibn-SS2025.pdf

5.1 Why synchronization is necessary

Threads belonging to the same process have:

- separate registers and stacks;
- a shared address space containing global variables, heap objects and other shared resources.

Sharing makes communication inexpensive, but it allows multiple threads to access the same data concurrently.

Consider:

```
counter = counter + 1
```

Although this looks like one operation, it normally consists conceptually of:

```
LOAD  counter into register R
ADD   1 to R
STORE R into counter
```

A thread can be interrupted between any two of these instructions.

Suppose `counter = 0` and two threads both increment it:

Step	Thread A	Thread B	Shared <code>counter</code>
1	Load <code>counter</code> into RA (RA=0)		0
2		Load <code>counter</code> into RB (RB=0)	0
3	Add 1 to RA (RA=1)		0
4	Store RA		1
5		Add 1 to RB (RB=1)	1
6		Store RB	1

The correct result would be 2, but the final result is 1. One increment has been overwritten: a **lost update**.

This can happen on a single-core CPU. The threads do not need to execute physically at the same instant; preemption and interleaving are enough.

5.2 Race conditions

A **race condition** exists when the correctness or result of a program unintentionally depends on the timing or execution order of concurrent operations.

A typical race condition requires:

1. shared state;
2. concurrent accesses to that state;
3. at least one access that modifies it;
4. insufficient synchronization.

The output may differ between executions because the scheduler can produce different interleavings.

Critical region

A **critical region** or **critical section** is a part of a program that accesses shared data and must not be executed simultaneously with a conflicting critical region.

```
entry protocol

critical region

exit protocol

non-critical region
```

The entry and exit protocols are implemented using a mutex, semaphore or another synchronization primitive.

Requirements of a good mutual-exclusion solution

A correct solution should provide:

- **Mutual exclusion:** conflicting threads are never simultaneously inside their critical regions.
- **Progress:** a thread outside its critical region must not unnecessarily prevent others from entering.
- **No timing assumptions:** correctness must not depend on CPU speed or the number of cores.
- **No starvation/bounded waiting:** a thread requesting entry should not be postponed forever.

The first is a safety property: “nothing bad happens.” Progress and absence of starvation are liveness properties: “something good eventually happens.”

5.3 Atomicity

An operation is **atomic** if other threads cannot observe or interrupt a partially completed execution of that operation.

For example, the abstract operation

```
wait(S)
```

must itself be atomic. If two threads could simultaneously test and decrement the semaphore non-atomically, the semaphore would not provide synchronization.

Protecting a critical region does not mean that all its ordinary instructions become individually atomic. It means that the entry protocol ensures that only the permitted thread or threads can execute that region.

5.4 Semaphores

A semaphore contains:

- a nonnegative count representing available permits or resource instances;
- an associated queue of blocked threads;

- two atomic operations, conventionally called **wait** and **signal**.

Other common names are:

Operation	Alternative names
wait(S)	P(S), down(S), sem_wait(S)
signal(S)	V(S), up(S), post(S), sem_post(S)

wait(S)

Conceptually:

```
if a permit is available:
    consume one permit
    continue
else:
    block until a permit becomes available
```

Thus, **wait** may block.

signal(S)

Conceptually:

```
make one permit available
if a thread is waiting:
    wake one waiting thread
```

signal does not normally block.

The implementation may represent blocked threads using negative values or separate queues, but exam solutions should normally describe the abstract behaviour rather than depend on a particular representation.

Binary and counting semaphores

A **binary semaphore** has at most one available permit:

```
Semaphore mutex = 1
```

A **counting semaphore** can represent several identical resource instances:

```
Semaphore printers = 3
```

Up to three successful `wait(printers)` operations can occur before another thread must release a permit.

5.5 Mutual exclusion with a semaphore

Initialize the semaphore to 1:

```
Semaphore mutex = 1
```

Every thread uses:

```
wait(mutex)

    // critical region

signal(mutex)
```

The first thread consumes the only permit. Other threads calling `wait(mutex)` must block until the first thread calls `signal(mutex)`.

The protected operation must lie entirely between `wait` and `signal`:

```
wait(mutex)
counter = counter + 1
signal(mutex)
```

Protecting only the load or only the store would still allow an unsafe interleaving.

Mutex versus binary semaphore

At the level required by many introductory exercises, a mutex can be viewed as a binary lock. More precisely:

- a mutex normally has an owner;
- only the thread that locked it should unlock it;
- a semaphore represents permits and does not necessarily have ownership.

Semaphores can consequently express both mutual exclusion and dependencies between different threads.

5.6 Enforcing an execution order

Suppose block X in Thread 1 may execute only after block Y in Thread 2.

Initialize a semaphore to zero:

```
Semaphore finished_Y = 0
```

Thread 1:

```
wait(finished_Y)
X
```

Thread 2:

```
Y
signal(finished_Y)
```

Why initialize it to zero?

- If Thread 1 reaches `wait` first, it blocks.
- After Thread 2 completes Y, `signal` provides the permit.
- If Thread 2 finishes first, the semaphore retains the permit. Thread 1's later `wait` consumes it and continues immediately.

This is what is meant by saying that a semaphore has **memory**: an earlier signal can be remembered in its counter.

5.7 Blocking versus busy waiting

A primitive can wait in two ways.

Busy waiting

```
while lock is unavailable:  
    repeatedly test lock
```

The thread remains runnable and consumes CPU time. A lock implemented this way is often called a **spinlock**.

Busy waiting can be reasonable for extremely short waits inside kernels or multiprocessor code, but it is inefficient for long waits.

Blocking

If the resource is unavailable:

1. the thread is placed in a waiting queue;
2. its state becomes **Waiting/Blocked**;
3. another runnable thread receives the CPU;
4. a later **signal** makes the blocked thread **Ready**.

Notice that waking a thread normally produces:

```
Waiting → Ready
```

not immediately **Waiting → Running**.

5.8 Condition variables

A condition variable allows a thread to sleep until some condition involving shared state may have become true.

It must be used together with:

- a **predicate**, such as `count > 0` or `production_done`;
- a mutex protecting that predicate;
- the condition variable on which threads wait.

Canonical pattern:

```
lock(mutex)

while (!condition) {
    cond_wait(condition_variable, mutex)
}

// condition is true; access shared state

unlock(mutex)
```

`cond_wait(cv, mutex)` performs three important actions:

1. atomically releases `mutex` and puts the caller to sleep;
2. waits for a notification;
3. reacquires `mutex` before returning.

The atomic release-and-sleep step prevents the following lost-wakeup race:

1. waiter checks that the condition is false;
2. notifier makes it true and sends a signal;
3. waiter goes to sleep after the signal;
4. no future signal arrives.

Why `while`, not `if`?

After waking, the thread must check the predicate again because:

- wakeups may be spurious;
- another awakened thread may acquire the mutex first and consume the resource;
- a notification means “the condition may now be true,” not “the condition is guaranteed to remain true.”

Correct:

```
while (count == 0)
    cond_wait(not_empty, mutex)
```

Unsafe:

```
if (count == 0)
    cond_wait(not_empty, mutex)
```

signal versus broadcast

- `cond_signal(cv)` wakes at least one waiting thread.
- `cond_broadcast(cv)` wakes all threads waiting on `cv`.

A producer adding one item normally needs to wake only one consumer. If `production_done` becomes true and every consumer must notice it and terminate, a broadcast is appropriate.

The awakened threads do not all enter the critical region simultaneously: each must first reacquire the mutex.

Condition variables do not remember notifications

If `cond_signal` or `cond_broadcast` is called while nobody is waiting, the notification disappears.

This differs from a semaphore:

```
signal(S)
```

with no waiter increments or preserves the semaphore's permit count, so a later `wait(S)` can consume it.

A mutex remembers whether it is currently locked, but an `unlock` is not a stored notification that a future thread can consume.

Why update and signal under the mutex?

The standard protocol is:

```
lock(mutex)
done = true
cond_broadcast(done_changed)
unlock(mutex)
```

The mutex makes changing the predicate and notifying waiters part of one synchronized protocol. A waiter cannot slip between checking `done` and beginning to wait. When it wakes and reacquires the mutex, it sees a consistent value of `done`.

Example: producer-consumer queue

- producers create items / jobs and insert them into a shared queue.
- consumers remove items
- a consumer cannot remove an item while the queue is empty
- instead of repeatedly checking the queue in a busy loop, it sleeps on `not_empty`

Concrete Example: downloading and playing a video

The producer downloads video chunks:

```
network -> buffer
```

The consumer decodes and plays them

```
buffer -> video player
```

1. the downloader puts some chunks into the buffer
2. the player begins consuming them
3. both continue concurrently

Other canonical examples:

- a network interface receives packets, while an application processes them
- a web server accepts requests while worker threads handle earlier requests
- one thread reads files, while another processes their contents.
- keyboard input arriving while the application processes earlier input

The general pattern is that

a buffer creates a pipeline and absorbs temporary differences in speed between producers and consumers

Benefits of concurrent production and consumption

1. **lower latency:** the consumer can start as soon as the first item exists instead of waiting for all items
2. **higher throughput:** production and consumption may overlap
 1. one core produces while another consumes
 2. a network or disk operation runs while the CPU processes previous data

3. **bounded memory use:** for a large or infinite stream, we may use a bounded buffer:

```
producer -> fixed-size buffer -> consumer
```

A bounded buffer typically uses two predicates and two condition variables:

```
std::condition_variable not_empty; // consumers wait here
std::condition_variable not_full;  // producers wait here
```

Complete producer-consumer program

```
shared Queue jobs
shared Mutex mutex
shared ConditionVariable not_empty
shared Boolean production_done = false

function producer():
    for number from 1 to 10:
        job = create_job(number)

        lock(mutex) //begin critical region 1

        jobs.push(job)
        signal(not_empty)

        unlock(mutex)

        // end critical region 1

    // no more jobs will be produced
    lock(mutex) // begin critical region 2

    production_done = true
    broadcast(not_empty)

    unlock(mutex) // end cr 2

function consumer(worker_id):
    while true:
        lock(mutex)

        while jobs.empty() and not production_done:
```

```

        wait(not_empty, mutex)

    if jobs.empty() and production_done:
        unlock(mutex)
        return

    job = jobs.pop()

    unlock(mutex)

    // processing is done outside of critical region
    process(job)

function main():
    producer_thread = start_thread(producer)
    consumer_thread_1 = start_thread(consumer, 1)
    consumer_thread_2 = start_thread(consumer, 2)

    join(producer_thread)
    join(consumer_thread_1)
    join(consumer_thread_2)

    print("all jobs completed")

```

Bounded-buffer producer-consumer problem

uses a shared queue with a **fixed maximum capacity**

- consumers wait on `not_empty` when the buffer is empty
- producers wait on `not_full` when the buffer is full
- a mutex protects the buffer and related shared state

The bounded buffer provides **backpressure**: if consumers cannot keep up, producers are forced to wait instead of allowing the queue to grow indefinitely.

Practical Example

Consider a program that reads large images from disk and processes them:

- the producer reads images and places them in memory
- several consumers resize / transform or analyze the images

- each image may occupy considerable memory

An unbounded queue could consumer all available memory if disk reading is temporarily faster than image processing. A buffer limited to, say, five images ensures that at most five unprocessed images are held in memory

Complete typed pseudocode

```
// shared state

constant BUFFER_CAPACITY = 5
constant NUMBER_OF_JOBS = 20
constant NUMBER_OF_CONSUMERS = 2

shared Queue<Job> jobs

shared Mutex mutex
shared ConditionVariable not_empty
shared ConditionVariable not_full

shared production_done = false

// producer
function producer() -> Void:
    for job_number from 1 to NUMBER_OF_JOBS:
        job = new Job(job_number)

        lock(mutex)

        while jobs.size() == BUFFER_CAPACITY:
            wait(not_full, mutex)

        // buffer is not full
        jobs.push(job)

        // buffer is not empty
        signal(not_empty)

        unlock(mutex)
```

```

    lock(mutex)

    production_done = true
    // every sleeping consumer must discover that production is finished
    broadcast(not_empty)

    unlock(mutex)

// consumer

function consumer() -> Void:
    while true:
        lock(mutex)

        while jobs.empty() and not production_done:
            wait(not_empty, mutex)

        // no buffered jobs remain, and no new jobs will arrive
        if jobs.empty() and production_done:
            unlock(mutex)
            return

        // buffer not empty
        job = jobs.pop()

        // at least one producer may now be able to proceed
        signal(not_full)

        unlock(mutex)

        process(job)

// main program

function main() -> Void:
    consumers = new Array<Thread>(NUMBER_OF_CONSUMERS)

    for i from 0 to NUMBER_OF_CONSUMERS - 1:
        consumers[i] = start_thread(consumer)

    producer_thread = start_thread(producer)

```

```

join(producer_thread)

for i in [1, NUMBER_OF_CONSUMERS]:
    join(consumers[i])

print("all jobs completed")

```

i join()

join() does not end the thread. It makes the calling thread wait until another thread has finished.

```

consumer_thread_1 = start_thread(consume, 1)

...

join(consumer_thread_1)

```

1. start a new hthread executing `consumer(1)`
2. continue executing the main thread
3. When `join(consumer_thread_1)` is reached, pause the main thread until `consumer_thread_1` terminates

This guarantees that the main thread will not prematurely terminate before `consumer_thread_1` does, which would be possible because the scheduler can decide to execute the return or the final statement of main, ending the main thread, before `consumer_thread_1` is finished

5.9 Producer–consumer pattern

For a bounded buffer of capacity (N), a semaphore solution uses:

```

Semaphore empty = N
Semaphore full  = 0
Semaphore mutex = 1

```

Producer:

```
produce(item)

wait(empty)
wait(mutex)
insert(item)
signal(mutex)
signal(full)
```

Consumer:

```
wait(full)
wait(mutex)
item = remove()
signal(mutex)
signal(empty)

consume(item)
```

Interpretation:

- **empty** counts free slots;
- **full** counts stored items;
- **mutex** protects the buffer data structure.

Do not acquire **mutex** before waiting for **empty** or **full**. Otherwise a thread might sleep while holding the buffer lock, preventing the other side from changing the condition it is waiting for.

5.10 Simplified read–write lock

Suppose:

```
Semaphore S = N
```

A reader consumes one permit:

```
lock_read:
    wait(S)

unlock_read:
    signal(S)
```

A writer attempts to consume all (N) permits:

```
lock_write:
    repeat N times:
        wait(S)

unlock_write:
    repeat N times:
        signal(S)
```

This appears to permit either:

- up to (N) simultaneous readers; or
- one writer holding every permit.

But `lock_write` is not atomic.

For (N=3), two writers could execute:

```
W1: wait(S)    // S: 3 → 2
W2: wait(S)    // S: 2 → 1
W1: wait(S)    // S: 1 → 0
W2: wait(S)    // blocks
W1: wait(S)    // blocks
```

Neither writer can acquire all three permits, and neither can release its partial acquisition because it has not entered the critical region. This is a **deadlock**.

A simple repair serializes writers while they collect the permits:

```
Semaphore S = N
Semaphore writer_gate = 1

lock_write:
    wait(writer_gate)
    repeat N times:
        wait(S)
    signal(writer_gate)

unlock_write:
    repeat N times:
        signal(S)
```

Only one writer may now attempt the multi-step acquisition.

This repair prevents the writer–writer split deadlock, but it does not necessarily guarantee fairness: continuously arriving readers might still starve a writer.

One correction to retain from your earlier work: your old Sheet 3 answer identified the interleaved writer acquisition, but described it as permitting simultaneous writing. The direct consequence is instead the deadlock shown above. 14-Operating-Systems-and-Networks-SoSe-25-Solutions.pdf

5.11 Common exam traps

- Treating `counter++` as an indivisible operation.
- Thinking a race requires multiple CPU cores.
- Protecting different threads with different mutexes.
- Forgetting to initialize a mutex semaphore to 1.
- Initializing an ordering semaphore to 1 instead of 0.
- Forgetting that semaphore operations must themselves be atomic.
- Omitting `signal` on some control-flow path.
- Sleeping while holding a mutex needed by the thread that could wake you.
- Using `if` instead of `while` around `cond_wait`.
- Assuming condition-variable notifications are remembered.
- Confusing deadlock with starvation.
- Calling every timing-dependent output a race: nondeterministic ordering can be intentional and harmless.

5.12 Exam-like exercises

Do these closed-book. Allow approximately 35–45 minutes in total.

Exercise 1: Race condition and semaphore repair

Two threads share:

```
integer counter = 0
```

Thread A:

```
repeat 3 times:
    counter = counter + 1
```

Thread B:

```
repeat 2 times:
    counter = counter + 1
```

Assume a single-core system and that an increment consists of:

```
LOAD counter into private register R
ADD 1 to R
STORE R into counter
```

Questions

1. Define the term **race condition**.
2. What should the final value of **counter** be?
3. Give an instruction-level interleaving in which one update is lost.
4. Explain why the race can occur despite there being only one CPU core.
5. Repair the program using one semaphore. State its initial value.
6. Explain completely but concisely what **wait** and **signal** do.
7. Would it be sufficient for A and B to use two different mutexes? Explain.

Use this table for Part 3:

Step	Thread	Instruction	Private register	Shared counter
1				
2				

Exercise 2: Ordering and condition variables

Part A: Ordering

Block `send_result` in Thread A may execute only after Thread B has completed `calculate_result`.

Write the simplest semaphore-based pseudocode for both threads. Give the semaphore's initial value.

Your solution must work regardless of which thread starts first.

Part B: Find the condition-variable errors

A consumer uses:

```
lock(mutex)

if (count == 0)
    cond_wait(not_empty, mutex)

item = remove_from_buffer()
count = count - 1

unlock(mutex)
```

The producer uses:

```
add_to_buffer(item)
count = count + 1
cond_signal(not_empty)
```

Answer:

1. Identify at least two synchronization errors.
2. Rewrite the relevant consumer and producer pseudocode correctly.
3. Why must the consumer test the condition in a **while** loop?
4. What happens if `cond_signal(not_empty)` is called while nobody is waiting?
5. What would happen in the analogous situation with `signal(S)` on a semaphore?
6. When would **broadcast** be preferable to **signal**?

Exercise 3: Debugging a semaphore construction

Let (N=4):

```
Semaphore S = 4

lock_read:
    wait(S)

unlock_read:
    signal(S)
```

```
lock_write:
    for i = 1 to 4:
        wait(S)

unlock_write:
    for i = 1 to 4:
        signal(S)
```

1. Give an exact execution involving two writers that leads to failure.
2. Classify the failure as:
 - mutual-exclusion violation,
 - deadlock, or
 - starvation.
3. Explain precisely why neither writer can continue.
4. Repair the construction using one additional semaphore or mutex.
5. Does your repair also guarantee that a writer can never starve? Explain briefly.

5.13 Answers

Exercise 1